Studies Regarding the Ensquared Energy of a Geostationary Hyperspectral Infrared Sounder

Zhenglong Li, Timothy J. Schmit, Jinlong Li, Jun Li, and Andrew K. Heidinger

***Abstract*— Compared with low Earth orbit (LEO) satellites, the high altitude from geostationary Earth orbit (GEO) satellites leads to increased diffraction effects on hyperspectral infrared (HIR) sounders, which reduces the ensquared energy (EE) within the satellites' field of view (FOV) and increases the pseudo noise of the measurements. To help understand how the instrument performance is affected by EE for the Geostationary Extended Observations Sounder (GXS), a point spread function (PSF) is used to simulate the contribution of each location within and outside of an FOV (4 × 4 km at nadir). The PSF is applied to the Moderate Resolution Imaging Spectroradiometer (MODIS) airborne simulator (MAS) data with a spatial resolution of 50 m, to determine an appropriate EE for GXS. Although wavenumber dependent, an EE of 70% is recommended, which ensures that all GXS channels have pseudo noise less than the instrument specifications. Regardless of the EE value, the pseudo noise reduces the precision of the temperature and moisture sounding retrievals in the troposphere. Even with an EE of 70%, the pseudo noise slightly increases the root-mean-square error (RMSE) by 3%–4% for temperature and by 1%–4% for relative humidity. If an EE of 70% is difficult to meet, due to cost for example, a lower EE can be a good tradeoff with only a slight degradation in the sounding retrieval quality, which may be overcome with spatial averaging using the inverted cone method. An EE of 50% would lead to an RMSE increase of about 6% for temperature and 3%–6% for relative humidity.**

***Index Terms*— Ensquared energy (EE), geostationary extended observations (GeoXO), geostationary extended observations sounder (GXS), hyperspectral infrared (HIR) sounder, point spread function (PSF), pseudo noise.**

I. Introduction

HYPERSPECTRAL infrared (HIR) sounders [1] from low Earth orbits (LEO), such as the atmospheric infrared sounder (AIRS), the cross-track Infrared sounder (CrIS), and infrared atmospheric sounding interferometer (IASI), have played a significant role in reducing forecast errors in global numerical weather prediction (NWP) models [2], [3], [4] from direct radiance assimilation. LEO HIR sounders are also widely used for nowcasting [5] and atmospheric composition studies [6]. One significant limitation of LEO HIR sounders is the lack of temporal resolution. LEO typically has a revisit frequency of twice a day for tropics and mid-latitudes. Even combining different existing LEO HIR sounders, the revisit rate is still inadequate. For example, IASI is in a morning orbit (9:30 A.M.), CrIS is in an afternoon orbit (1:30 P.M.), and the recently launched Fengyun-3E is in an early morning orbit (5:30 A.M.). These LEO HIR sounders, although with global coverage, have a gap of 4 h between them. The low revisit rate makes LEO HIR sounders less effective for monitoring and providing thermodynamic information for fast changing high-impact weather events, such as local severe storms. Currently, forecasters heavily rely on geostationary Earth orbit (GEO) imagers to monitor the development of such high-impact weather events. Due to limited information content from imagers with limited spectral bands, GEO imagers can only provide limited thermodynamic information [7], [8], resulting in a lack of vertical resolution. GEO HIR sounders can fill the gap left by LEO HIR sounders and GEO imagers [9], and they can provide a unique value for depicting the diurnal variation in various applications [10], such as low-level moisture [11], land surface emissivity [12], and atmospheric trace gases [6], [13]. Other applications that can benefit from the high temporal resolution from GEO HIR include atmospheric motion vector [14], monitoring and nowcasting short-term local severe storms [15], and so on.

Compared with LEO, GEO HIR sounders are more challenging technically, due to the much higher satellite altitude of 35 700 km. At this height, the diffraction effects for a GEO HIR sounder become more prominent than for LEO (as shown in Section II-A), although a three-axis satellite platform in the GEO orbit allows for much longer dwell or look times. The diffraction happens naturally when the light travels through a satellite sensor's aperture and is spread out. The diffraction effects may lead to lights from the field of view (FOV) of interest not reaching the detector and lights from outside of the FOV reaching the detector, therefore reducing the ensquared energy (EE) from the FOV and increasing the energy from outside of the FOV. EE is defined as the percentage of energy coming from the FOV of interest. A high percentage of EE is needed in order to maintain the high representativeness of the FOV and the usefulness of GEO HIR sounder measurements. The ideal EE is when all the energy coming from the FOV

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Zhenglong Li and Jinlong Li are with the Cooperative Institute for Meteorological Satellite Studies, Space Science and Engineering Center, University of Wisconsin–Madison, Madison, WI 53706 USA (e-mail: Zhenglong.li@ssec.wisc.edu).

Timothy J. Schmit is with the NOAA/Center for Satellite Applications and Research, Madison, WI 53706 USA.

Jun Li is with the National Satellite Meteorological Center, China Meteorological Administration, Beijing 100081, China (e-mail: junli@cma.gov.cn).

Andrew K. Heidinger is with the GeoXO Program Office, NESDIS/NOAA, Madison, WI 53706 USA.

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of interest has an even weight everywhere. A higher EE can be realized by increasing the aperture diameter of the sensor. However, a larger aperture diameter increases the cost of manufacturing the sensor as well as the cost to get the sensor into the orbit. Thus, a reasonable EE, which still ensures the usefulness of measurements, must be scientifically determined in order to maintain a cost-effective sensor.

NOAA is planning for the next generation of GEO satellites through the geostationary extended observations (GeoXO) program. An HIR sounder, called the GeoXO Sounder or Geostationary Extended Observations Sounder (GXS), is planned for a central satellite location around 105°W. GXS will have two spectral bands, the longwave band of 680–1095 cm^{-1} (or 9.13–14.71 μm) and the midwave band of 1689–2250 cm^{-1} (or 4.44–5.92 μm), with a spectral resolution of 0.625 cm^{-1} and a spatial resolution of 4 km at nadir view [16]. GXS will be particularly useful for NWP modeling, nowcasting, and atmospheric composition, among many other areas. Zhang et al. [17] used the Moderate Resolution Imaging Spectroradiometer (MODIS) airborne simulator (MAS) data (with a spatial resolution of 50 m) to investigate how different EE values affect the measurements' representativeness. Using methodologies developed by the Cooperative Institute for Meteorological Satellite Studies (CIMSS) at the University of Wisconsin–Madison, Grandell and Stuhlmann [18], [19] studied the impact of diffraction on the infrared sounder (IRS) on the meteosat third generation (MTG), which will be launched in the coming years. They used AIRS data to quantify the impact on radiance measurements and temperature retrievals. According to [20], the IRS will have a spectral mean EE greater than 66.0% for the longwave band and greater than 71.8% for the midwave band.

While EE itself is not related to instrument noise, the impact of EE has to be quantified and compared with the instrument's noise. Thus, each GEO HIR with a different instrument noise may require a different EE. Determining the appropriate EE is an important step in the planning and development of new instruments. Similar studies have been carried out for GOES-R [17] and MTG [18], [19]. In this study, MAS data are used to understand the impact of the diffraction effects on measurements and to determine an appropriate EE for GXS.

Section II introduces the mathematical definition of EE. Section III presents the simulation studies on how EE affects the measurements using MAS data. The impact of EE on atmospheric sounding retrievals is presented in Section IV. A summary and discussion are provided in Section V.

II. Ensquared Energy

EE characterizes how much energy is coming from the FOV of interest. A larger EE is preferred to be more representative of the FOV. EE can be estimated using the system point spread function (PSF), which is defined as the response of a focused satellite sensor to a point source. Following previous work [18], [19], [20], the system PSF is determined by two PSFs: the diffraction PSF and the detector PSF. The diffraction PSF is the sensor's response to a point source due to diffraction effects without consideration of the sensor's footprint size.

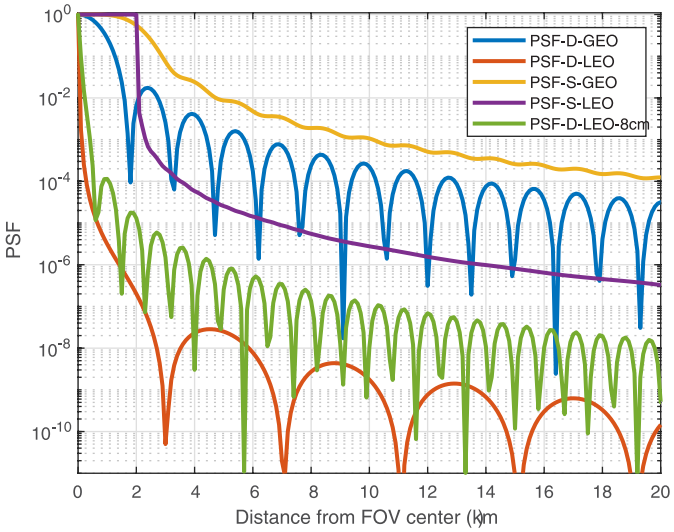


Fig. 1. PSFs due to diffraction effects as a function of distance from the FOV center for GEO (blue) and LEO (red) as well as the system PSF for GEO (yellow) and LEO (purple). A wavelength of 11 μm and an aperture diameter of 27 cm are used. The green line shows PSFs due to diffraction effects for LEO with an aperture diameter of 8 cm.

Also, the detector PSF is the sensor's response to the scene without diffraction effects for a specific footprint size.

A. Diffraction PSF

The diffraction PSF explains how diffraction affects the measurements. It is commonly simulated using [18], [19], [20]

$$\mathrm{PSF}_{\mathrm{dif}} = \left[\frac{2J_1\left(\frac{\pi D\sqrt{x^2+y^2}}{\lambda H}\right)}{\frac{\pi D\sqrt{x^2+y^2}}{\lambda H}} \right]^2 \tag{1}$$

where J_1 is the first-order Bessel function of the first kind, D is the aperture diameter, x and y are the distances from the center of the FOV in two directions, λ is the wavelength, and H is the satellite altitude.

Fig. 1 shows diffraction PSFs for a wavelength of 11 μm and an aperture diameter of 27 cm (same as the Advanced Baseline Imager on GOES-R). The impact of diffraction effects on GEO is much more evident than that on an LEO. For example, the FOV center has a normalized PSF of 0.37% for GEO, which means that 0.37% of the energy is coming from the center of a small area of 0.1 × 0.1 km. On the contrary, an LEO with the same aperture diameter of 27 cm has 98.67% of the energy from the same area, which means that the diffraction effects on LEO are much weaker. Therefore, LEO sensors can have small aperture diameters while maintaining a high EE (i.e., the CrIS aperture is 8 cm). When the LEO aperture diameter is reduced to 8 cm, the diffraction effects become stronger. However, 61.05% of the energy still comes from the same center area, again indicating that the diffraction effects on LEO are much weaker and are practically negligible. The diffraction effects must be quantified in order to understand the representativeness of GEO measurements. More specifically, the uncertainties in the measurements introduced due to the diffraction effects must be understood.

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B. System PSF

Once the diffraction PSF is known, the system PSF can be simulated using

$$\mathrm{PSF}_{\mathrm{sys}} = \frac{\mathrm{PSF}_{\mathrm{det}} \otimes \mathrm{PSF}_{\mathrm{dif}}}{\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \mathrm{PSF}_{\mathrm{det}} \otimes PSF_{\mathrm{dif}} dxdy} \tag{2}$$

where $\mathrm{PSF}_{\mathrm{sys}}$ is the system PSF after convoluting with the detector PSF or $\mathrm{PSF}_{\mathrm{det}}$ and $\otimes$ denotes the convolution. Also, the detector PSF can be simulated using a simple boxcar function

$$\mathrm{PSF}_{\mathrm{det}} = \begin{cases} 1, & \text{if} |x| < \frac{1}{2}d \text{ and } |y| < \frac{1}{2}d \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where d is the FOV size (4 × 4 km for GXS). Equation (3) indicates that the detector has the full response to the scene within the FOV and no response outside the FOV when no diffraction effects are considered. Convolving the diffraction PSF with the detector PSF to obtain the system PSF is not considered in [17], which results in an overestimate of EE. Fig. 1 shows that the GEO system PSF is much more spread out than the diffraction PSF, although it appears that most energy is coming from an area less than 2 km from the FOV center. Fig. 1 also shows that an LEO HIR sounder with the same aperture diameter of 27 cm will have a system PSF very similar to the boxcar-shaped detector PSF. This again confirms that the diffraction effects are negligible for an LEO HIR sounder. Mathematically, EE is defined as

$$\mathrm{EE} = \frac{\int_{-\frac{1}{2}d}^{+\frac{1}{2}d}\int_{-\frac{1}{2}d}^{+\frac{1}{2}d} \mathrm{PSF}_{\mathrm{sys}} dxdy}{\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \mathrm{PSF}_{\mathrm{sys}} dxdy}. \tag{4}$$

The EE shows how much energy is coming from the FOV. For the wavelength of 11 μm and the aperture diameter of 27 cm, the EE is 69% for GXS.

Fig. 2 shows the EE as a function of aperture diameters for selected GXS wavenumbers. These four channels are the starting and ending channels of longwave and midwave spectral bands of GXS. It can be seen that the EE is significantly affected by the aperture diameters. Larger diameters lead to larger EE, but the relationship is not linear. For apertures with diameters larger than 40 cm, the benefit of further increasing the diameters is not as significant as those smaller than 40 cm. Thus, a reasonable EE should be set as higher EE values will significantly increase the cost.

Fig. 2 also shows that EE is highly related to wavelengths. Long wavelength (small wavenumber) is much more affected by EE than shorter wavelengths. For example, an aperture diameter of 20 cm leads to an EE of 80% for 2250 cm^{-1}, while for 1689, 1095, and 680 cm^{-1}, the EE reduces to 76%, 67%, and 54%, respectively. This is apparently because longer wavelengths are more prone to diffractions. These results indicate that the impact of the aperture diameter on EE is less of a concern for the midwave spectral band than the longwave spectral band. Therefore, an EE should be recommended based on the analysis of the longwave spectral band. In this study, the EE is determined based on the longest wavelength of 680 cm^{-1} for GXS.

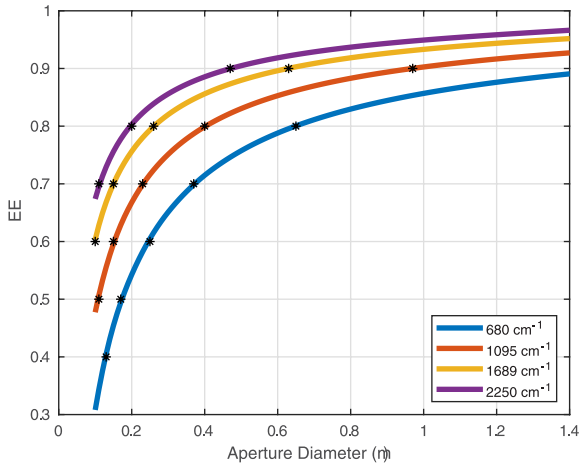


Fig. 2. EE as a function of the aperture diameter for selected wavenumbers of GXS.

Note that using the system PSF from (2) only accounts for the detector and diffraction effects. The EE is also affected by other factors such as the optical image quality with a diffraction-limited system, the vibration from a cryocooler, the motion of the line of sight due to the spacecraft motion during the data acquisition time, and the electrical crosstalk. These factors likely reduce the EE as well.

III. Determination of EE

EE should neither be too large due to increased cost nor be too small due to reduced representativeness of the FOV. Quantitatively, the EE can be determined based on the comparison of the pseudo noise to the instrument noise requirement. The pseudo noise is defined as

$$\delta R = \sqrt{\overline{(R^o - R^{\mathrm{ideal}})^2}} \tag{5}$$

where δR is the pseudo noise, R^o is the observation without instrument noise, and R^{ideal} is the ideal observation if EE is 100% with even weights, also without instrument noise. The bar means the average. The observation R^o can be obtained through the convolution of the system PSF with the scene radiance using

$$R^o = \frac{\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \mathrm{PSF}_{\mathrm{sys}} \otimes R_s dxdy}{\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \mathrm{PSF}_{\mathrm{sys}} dxdy} \tag{6}$$

where R_s is the scene radiance at location (x, y). For the ideal observation, the ideal system PSF with a boxcar shape should be used in (6). Understandably, small pseudo noise requires large EEs, which in turn requires large aperture diameters and increased costs. In this study, an EE will be determined when the pseudo noise is comparable to the instrument noise.

A. MAS Data

Similar to [17], the MAS data will be used to simulate GXS observations using (6). MAS is an airborne scanning spectrometer onboard the NASA ER-2 aircraft [21]. At an altitude of 20 km, MAS provides radiance measurements with a spatial resolution of 50 m. This very fine spatial resolution makes MAS ideal to study the impact of EE on pseudo

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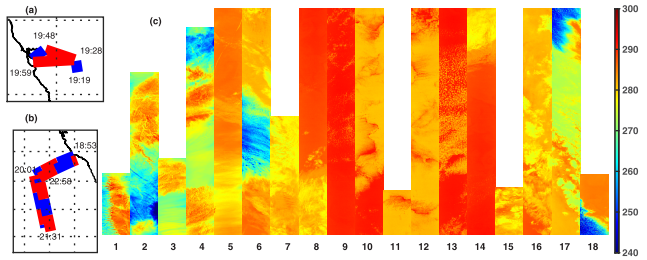


Fig. 3. Flight tracks of the MAS data used for (a) first flight with four tracks on January 31, 2003, and (b) second flight with 14 tracks on April 10, 2003. Consecutive flight tracks are shown in different colors. Also shown is the time of the flight at selected locations in UTC, which tells the flight direction. The inbound flight on April 10, 2003 has the exact same ground track as the outbound flight. It is manually shifted to show the different flight tracks. (c) 11.02-μm brightness temperature imagery from all 18 flight tracks. Numbers on the bottom show the flight track number. For each image, the flight direction is from the bottom to the top. Most of the regions with cold brightness temperature shown in yellow or blue are cloudy, such as those of flight tracks 5–18 from the second flight over ocean. Some cold brightness temperatures from the first flight are over mountains, such as the left part of flight track 1, the first half of flight track 2, and the last quarter of flight track 4.

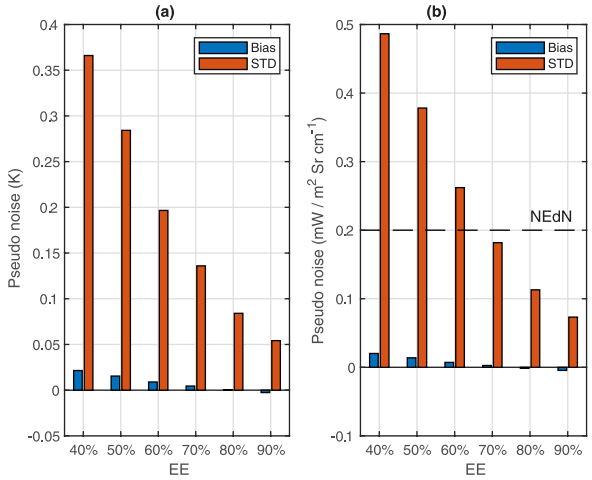


Fig. 4. Pseudo noise as a function of EE in (a) brightness temperature (K) and (b) radiances (mW m^{-2} Sr^{-1} [cm^{-1}]$^{-1}$). The EE numbers are determined based on the wavenumber of 680 cm^{-1}. NEdN represents the instrument noise equivalent radiance differential.

noise. Although there are 50 spectral bands from 0.445 to 14.428 μm, only 14 IR spectral bands that overlap the GXS spectrum are used.

Two flights from the Observing-system Research and Predictability Experiment were used (Fig. 3). The first flight took place in central California on January 31, 2003, from 19:19 to 20:24 UTC [Fig. 3(a)]. There are four flight tracks and 22 424 scanlines in total. The second flight took place in the eastern Pacific Ocean on April 10, 2003, from 18:53 to 23:58 UTC [Fig. 3(b)]. There are 14 flight tracks and 111 940 scanlines. Both flights have a significant percentage of cloud coverage as seen from the 11.02-μm brightness temperature imagery in Fig. 3(c). The first flight also has decent coverage of the mountains and valleys in flight tracks 1, 2, and 4, when flying over the Sierra Nevada Mountains. The mixture of land and ocean, clear and cloudy, mountains, and low elevations allows a meaningful evaluation of the impact of EE on the pseudo noise.

B. Evaluations Based on 11.02 μm

The evaluation is based on the comparison of pseudo noise with instrument noise. Fig. 4 shows the results from combining the two MAS flights for MAS band 45 centered at 11.02 μm. This band is highlighted for two reasons. First, as pointed out earlier, long wavelengths are more affected by EE than short wavelengths. Second, the clean IR window band is more sensitive to the surface and clouds, and thus, the scene should exhibit more heterogeneity than nonwindow spectral bands under most atmospheric conditions.

For both brightness temperatures and radiances, it is clear that a larger EE is beneficial because a lower pseudo noise is introduced. Both the mean bias and the standard deviations are reduced with increased EE. In particular, when the EE is 70% or larger, the pseudo noise in terms of standard deviation is smaller than the instrument noise equivalent differential radiance (NEdN, obtained from the official GXS Performance and Operation Requirements Document or PORD); when the EE is 60% or smaller, the pseudo noise standard deviation is larger than the instrument noise requirement. While the EE depends on wavelengths, the use of 11.02 μm represents a worse case scenario because this channel is more sensitive to the surface and clouds than other channels. Based on Fig. 4, an EE of 68% might meet the requirement that the pseudo noise is comparable to and smaller than the instrument noise. An EE of 68% corresponds to an aperture diameter of 34 cm for GXS.

The impact of EE on the pseudo noise is dependent on the homogeneity of the scene. For a homogeneous scene like open water, the scene radiance R_s will be the same everywhere. Also, (6) becomes

$$R^o = R_s. \tag{7}$$

This indicates that no matter what PSF is used, the observation is always the same. Therefore, the impact of EE on the pseudo noise is much less over a homogeneous scene. Fig. 5 shows the pseudo noise for the land and ocean flights separately. Because the ocean has a more homogeneous surface, it is much less affected by the EE. The EE can be reduced to 64% to ensure that the pseudo noise standard deviation is smaller than the instrument noise. On the other hand, an EE of 77% over land is needed to meet the same requirement. Similarly, clear scenes are a lot more homogeneous than cloudy ones. Thus, the cloudiness also has a significant impact on the pseudo noise, which is confirmed in Fig. 6. For clear scenes, the smallest EE of 58% can still ensure that the pseudo noise standard deviation is smaller than the instrument noise. On the other hand, for cloudy scenes, an EE of 70% is needed to meet the same requirement. A cloudy scene is determined when the 11.02-μm brightness temperature is colder than 285 K. This threshold is empirically determined based on the fact that low clouds over ocean have a brightness temperature of around 280 K in Fig. 3(c).

The pseudo noise bias is less affected by EE as its values are much smaller compared with standard deviation values. For smaller EE values, the bias is larger in absolute value. It is interesting that biases from cloudy and clear scenes have opposite signs: positive in cloudy scenes and negative in clear

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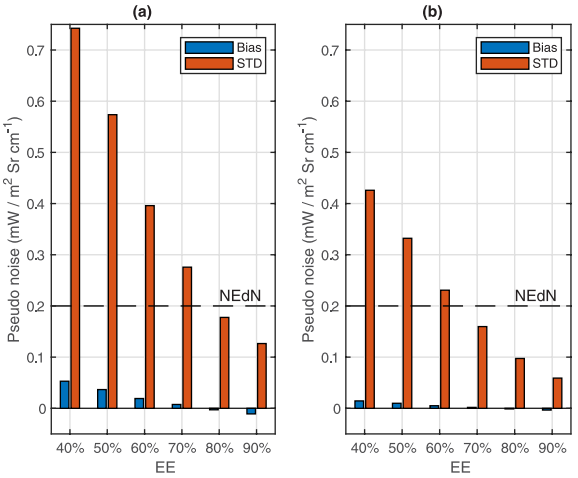


Fig. 5. Pseudo noise as a function of EE in radiances (mW m^{-2} Sr^{-1} [cm^{-1}]$^{-1}$) for (a) land and (b) ocean scenes. The EE numbers are determined based on the wavenumber of 680 cm^{-1}. NEdN represents the instrument noise equivalent radiance differential.

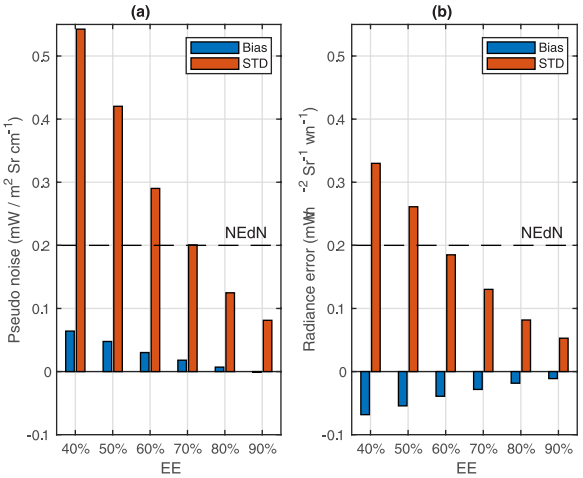


Fig. 6. Pseudo noise as a function of EE in radiances (mW m^{-2} Sr^{-1} [cm^{-1}]$^{-1}$) for (a) cloudy and (b) clear scenes. The EE numbers are determined based on the wavenumber of 680 cm^{-1}. NEdN represents the instrument noise equivalent radiance differential.

scenes. The scene outside of a cloudy FOV has three scenarios: clear, cloudy, or partially cloudy. Statistically, in any of these three scenarios, the radiance from outside a cloudy FOV will not be smaller and likely larger than the cloudy radiance. Contribution from outside of the cloudy FOV will make the observed radiance warmer than it should be, resulting in a positive bias. In clear scenes, it is the opposite, resulting in a negative bias. In fact, such a bias pattern is not necessarily inherent to a cloudy or clear scene. Any FOV that is colder (warmer) than its neighboring FOVs is likely to have a positive (negative) bias. This means that radiance measurements at the cloudy edge will likely have positive biases and at the clear edge will likely have a negative bias. This may make it more difficult for the cloud mask algorithm since those observations do not reflect the true state of the FOV.

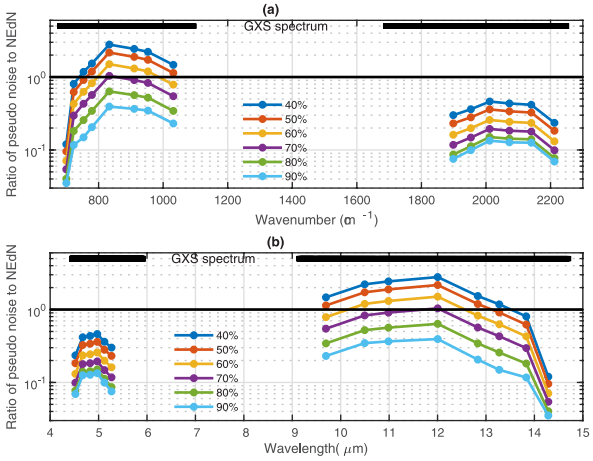


Fig. 7. Ratio of pseudo noise to the instrument noise equivalent radiance differential (NEdN) for different EE values. The EE numbers are determined based on the wavenumber of 680 cm^{-1}. (a) Wavenumber. (b) Wavelength.

C. EE Recommendation

The analysis in Section III-B is based on 11.02 μm. To find the EE that works for all GXS channels, a similar analysis was carried out for the 14 MAS channels that overlap with the GXS spectrum. Both land and ocean flights are used. Fig. 7 shows that the impact of EE on the GXS water vapor band is small. This is because the diffraction effects on shorter wavelengths are not as profound. Even with an EE of 40%, the maximum pseudo noise is only 46% of NEdN in this case, which is 4.97 μm or 2012 cm^{-1}. On both sides of 4.97 μm, the impact of EE becomes smaller. There is a strong CO_2 absorption band at 4.3 μm. To the left of 4.97 μm, as one gets closer to 4.3 μm, the CO_2 absorption becomes stronger, channels peak higher, and the temperature field becomes more homogeneous. This makes those channels less affected by EE. On the right side of 4.97 μm, there is a strong water vapor absorption band at 6.3 μm. This explains why the impact of EE becomes smaller as well. Thus, it is expected that the water vapor channels between 5.27 and 5.92 μm or 1897 and 1689 cm^{-1} are even less affected by EE.

As expected, longwave channels are strongly affected by EE. From Fig. 7, an EE of 70% is needed to ensure that all channels meet the requirement and that pseudo noise is comparable or smaller than NEdN. This corresponds to an aperture diameter of 37 cm. The impact of EE is strongest on 12.00 μm. Although 11.02 μm is more sensitive to the surface and clouds, 12.00 μm has a longer wavelength, making it more prone to the diffraction effects, thus more affected by EE. Similar to the water vapor channels, CO_2 absorption channels with wavelengths longer than 14.28 μm or 700 cm^{-1} will be less affected by EE than 14.28 μm. The same can be said for channels with wavelengths shorter than 9.7 μm or 1031 cm^{-1}.

It is understandable that an EE of 70% (or an aperture diameter of 37 cm) could be costly. A reduced EE of 60% can significantly reduce the aperture diameter to 25 cm, which is more cost effective. While such degradation in requirement will have a substantial impact on the instrument performance, the impact on the sounding channels might be small. As pointed out before, the EE impact on the water vapor band is small. Regardless of which EE is used, the pseudo noise introduced is always smaller than NEdN, even with an EE of 40%. With an EE of 60%, all the longwave CO_2 sounding channels longer than 12.5 μm will have a

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pseudo noise smaller than NEdN. Thus, reducing EE from 70% to 60% may result in limited degradation in atmospheric sounding information. On the other hand, the window channels between 10.2 and 12.5 μm will have pseudo noise larger than NEdN. The increased pseudo noise in IR window channels may reduce the sounding information in the lower troposphere, where those channels are sensitive.

IV. Impact Study on Sounding Retrievals

No matter which EE is used, the observed radiances will be affected by the pseudo noise, reducing the representativeness of the FOV that is observed. To understand how pseudo noise affects products, using sounding as an example, a simulation study is conducted for atmospheric profile retrieval.

The Seebor database [22] is used to simulate the IASI spectra using the stand-alone fast radiative transfer algorithm (SARTA) model [23]. All 15 704 profiles except those with latitudes outside of GXS domain are used. Each profile together with surface skin temperature and surface emissivity is used to simulate clear sky IASI radiances with different satellite zenith angles. The simulated IASI spectra are then converted to the GXS spectra [24] with Hamming apodization added. Random instrument noise based on GXS PORD is added. Additional random noise based on pseudo noise is added according to the ratios presented in Fig. 7. The ratios are linearly interpolated to the GXS spectrum. For GXS channels not covered by the MAS spectrum, linear extrapolation is performed with regard to the logarithm of ratios to ensure positive values. The random noise added has a Gaussian distribution with no bias. Thus, this simulation study does not address the impact of the biases discussed in Fig. 6(b).

A simple linear regression retrieval algorithm is used to retrieve temperature and moisture profiles from the GXS brightness temperature spectrum [25], [26]. The retrieval algorithm takes the form of

$$Y = KX \tag{8}$$

where Y is the vector of predictants, K is the regression coefficients, and X is the vector of the predictors. The predictors include GXS brightness temperatures that are simulated from the SeeBor database. The quadratic terms of brightness temperatures are also used to account for the nonlinearity. The brightness temperatures have random noise added based on the instrument's noise equivalent delta temperature and the pseudo noises. The predictants include the profile of temperature and moisture from the SeeBor database. The logarithm of the mixing ratio is used for an improved linear relationship between it and the brightness temperatures; 90% of data are used for training to generate the regression coefficients and the remaining is used for independent validation. Similar retrieval algorithms have been used to generate the first guess for physical retrievals for the GOES sounder [25]. Fig. 8 shows the impact of EE on temperature and moisture sounding retrievals. Regardless of which EE is used, when compared with ideal EE of 100%, a degradation of the profile retrieval in the whole troposphere is seen for both temperature and moisture. This degradation occurs because pseudo noise is introduced when

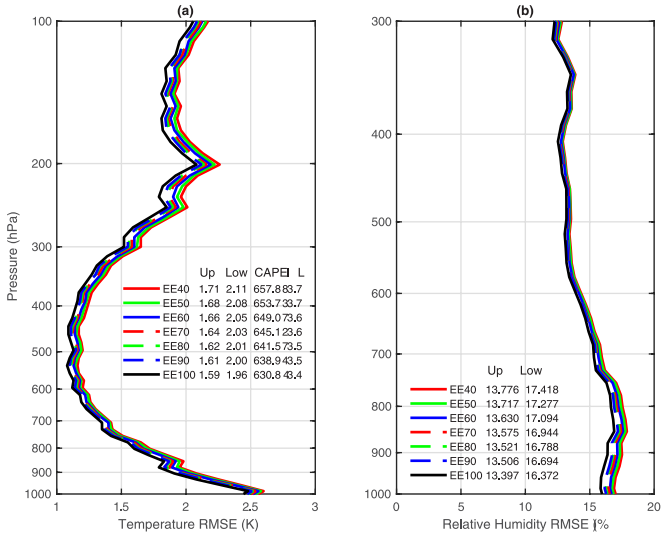


Fig. 8. RMSE of (a) temperature and (b) relative humidity with different pseudo noises. The level-averaged RMSE of up/low atmosphere temperature (K), the CAPE (J/Kg), and LI (K) is also shown. The upper atmosphere refers to 100–500 hPa for temperature and 300–700 hPa for moisture (%). The lower atmosphere refers to 800–1000 hPa. The retrieval relative humidity is calculated with the truth temperature to avoid uncertainty from the temperature.

TABLE I

RMSE Increase in Percentage of Different EEs for Different Retrieval Parameters Compared With the Ideal EE of 100%

EE (%)	Upper T	Lower T	CAPE	LI	Up RH	Low RH
40	7.6	7.7	4.3	9.9	2.8	6.4
50	5.7	6.1	3.6	8.4	2.4	5.5
60	4.4	4.6	2.9	6.7	1.7	4.4
70	3.1	3.6	2.3	5.2	1.3	3.5
80	1.9	2.6	1.7	3.8	0.9	2.5
90	1.3	2.0	1.3	2.9	0.8	2.0
100	0	0	0	0	0	0

EE is not 100%. Larger EEs introduce less pseudo noise, and the retrievals are less degraded.

Table I shows the root-mean-square error (RMSE) increase in percentage for different EE and different retrieval parameters. While an EE of 70% may have less pseudo noise than the sensor specification, the retrieval quality is still marginally affected. For temperature, the upper atmosphere has an RMSE increase of 3.1%, and the lower atmosphere has an RMSE increase of 3.6%. The RMSE of convective available potential energy (CAPE) and lifted index (LI) is increased by 2.3% and 5.2%, respectively. For relative humidity, the upper atmosphere has an RMSE increase of 1.3% and the lower atmosphere has an RMSE increase of 3.5%. When a smaller EE is used, introducing more energy from outside of an FOV, more degradation on the retrievals is seen. For example, an EE of 40% causes the temperature RMSE to increase substantially by 7.6% for the upper atmosphere and by 7.7% for the lower atmosphere; moisture RMSE increases marginally by 2.8% for the upper atmosphere and substantially by 6.4% for the lower atmosphere. These results indicate that the relative humidity retrievals are less affected by pseudo noise than the temperature. This is likely because the moisture channels are less affected by the EE and have less pseudo noise introduced than the longwave temperature channels in terms of the ratio

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to the sensor's specification between 13 and 14 μm, as seen from Fig. 7. Table I also shows that the lower atmosphere is more affected by EE than the upper atmosphere. This result occurs because the atmosphere is less homogeneous in the lower levels than the upper levels.

While a larger EE is preferred for better representativeness of the FOV, a small EE might be a good tradeoff with a slight degradation in the sounding retrievals. An EE of 50% would lead to an RMSE increase of about 6% for tropospheric temperature, less than 3% for the upper tropospheric relative humidity (300–700 hPa), and less than 6% for lower tropospheric relative humidity (800–1000 hPa). If the cost is the main limitation for the instrument development, one might choose to reduce the EE with a slight degradation in the sounding quality. More spatial averaging (e.g., for those channels with weighting functions peaking at the upper tropospheric layers) might alleviate this degradation with the inverted cone method [27].

This simulation study of atmospheric sounding retrievals should not be interpreted as an indication of the quality of GXS atmospheric sounding retrievals. This is due to the use of a simple linear regression technique. A more sophisticated retrieval algorithm, such as a physical retrieval algorithm [26], [28], [29] or machine learning-based algorithm [30], [31], should be used to maximumly explore sounding information from the GXS radiances. However, the RMSE increase due to reduced EE, especially in percentage, should be applicable to other retrieval algorithms as well.

It should be pointed out that the impact of EE due to the diffraction should not be confused with the refraction. The refraction happens when the light travels through two media with different refraction indices, causing bending of the light according to the famous Snell's law. The bending increases with the zenith angle and refraction indices. The refraction indices in the infrared, just like in the visible, increase with moisture and decrease with temperature due to the change in air density [32], [33]. The light bending introduces uncertainties in the geolocation. The uncertainties, according to [34], can be up to 2.22 and 17.85 m for zenith angles of 30° and 60°, respectively. For GXS with a footprint size of 4 km in nadir, such uncertainties can be practically ignored as the impact on the sounding retrievals is expected to be small.

V. Summary

NOAA is planning for the next generation of geostationary satellites with the GeoXO program, which is likely to carry an HIR sounder GXS. Due to the high altitude of geostationary satellites, the diffraction effects become inevitable, which cause substantial energy from outside of an FOV reaching the sensor, reducing the representativeness of it. To understand how the GXS radiance observations are affected, a PSF is used to simulate the diffraction effects. The EE, which characterizes the percentage of energy coming from an FOV, is studied.

The results show that larger aperture diameters reduce the diffraction effects and increase the EE, which is beneficial but also means more cost to build the instrument. The diffraction effects are more significant for long wavelengths than short ones. Because of the wavelength dependency, an EE should be provided with the wavelength specified. In this study for GXS, the EE is defined based on the longest wavelength (14.706 μm or 680 cm^{-1}). As a result, all other channels should have EE values larger than 14.706 μm. It is also confirmed that the diffraction effects on polar-orbiting HIR sounders are negligible.

When there is energy coming from outside of an FOV, the representativeness of the FOV is reduced. In ideal conditions, all energy should come from within the FOV with equal weight everywhere. Practically, that is not possible and results in the introduction of pseudo noise, which is the difference between the observed radiance and the ideal radiance. To understand the relationship between EE and pseudo noise, the MAS data with a spatial resolution of 50 m are used to simulate GXS observations with a square footprint size of 4 by 4 km. Two flights, one over land and the other over ocean, are used.

We found that pseudo noise increases rapidly with reduced EE. This is even worse for inhomogeneous scenes, i.e., cloudy scenes are more prone to pseudo noise than clear scenes and land scenes more prone to pseudo noise than ocean scenes. While 14.706 μm is more subject to diffraction effects, we found that 12 μm has more pseudo noise introduced than other channels. This is because 12 μm is sensitive to the surface, where the scene is more inhomogeneous than CO_2 channels. Based on the simulation study with MAS data, an EE of 70% with an aperture diameter of 37 cm is recommended, which ensures that all GXS channels have pseudo noise less than the instrument specifications.

To understand how the pseudo noise affects the atmospheric sounding retrievals, a simulation study is carried out with a simple linear regression algorithm. The pseudo noise is added on top of the instrument's noise equivalent delta temperature, and the root mean square of errors of temperature and relative humidity retrievals are compared with the ideal case where no pseudo noise is introduced. The results show that the pseudo noise degrades the retrieval quality of the whole profile in the troposphere for both temperature and moisture. However, the degradation is found to be weak. The temperature is found to be more affected by the pseudo noise than the moisture. This is likely because the longwave CO_2 channels between 13 and 14 μm are more affected by EE, which introduces more pseudo noise than the water vapor channels in the midwave band in terms of the ratio to the sensor's specification. Because the lower troposphere is less homogeneous than the upper troposphere, channels sensitive to the lower troposphere are slightly more affected than those sensitive to the upper troposphere, especially the moisture retrievals. If cost limitation is the main factor preventing the sensor's EE from reaching the recommended 70%, a smaller EE is probably a good tradeoff with only a slight degradation of the sounding quality.

The use of MAS data might underestimate the impact of EE because flights to take those measurements were usually taken in nonsevere weather conditions or more homogeneous scenes. While the spatial resolution of from MAS (50 m) is great for studying the diffraction effects, the limited spectral bands overlapping with GXS might have reduced representativeness for GXS spectral channels that are not well represented by MAS spectral bands, especially those on the absorption lines. This is because the Jacobians of those GXS channels usually peak higher than those nearby broadband MAS spectral bands.

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Since the atmosphere becomes more homogeneous away from the surface, those GXS channels likely see fewer spatial variations than nearby MAS spectral bands. As a result, the pseudo noise based on MAS spectral bands might be an overestimate of nearby GXS spectral channels that are on absorption lines. Thus, the impact of EE on pseudo noise and sounding retrievals might be overestimated for those absorption channels when using MAS data.

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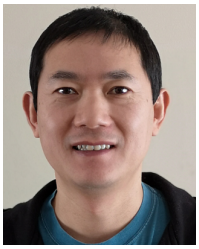
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Zhenglong Li received the B.S. and M.S. degrees in atmospheric sciences from Peking University, Beijing, China, in 2000 and 2003, respectively, and the Ph.D. degree in atmospheric sciences from the University of Wisconsin–Madison (UW), Madison, WI, USA, in 2009.

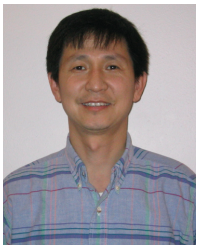
Since 2009, he has been a Research Scientist with the Space Science and Engineering Center, UW. He has more than 20 years of experience in meteorological satellite remote sensing, focusing on infrared sounders, and both traditional and advanced hyperspectral. His research interests include retrieval algorithm development for atmospheric profiles and land surface emissivity, product validation and evaluation, application to weather forecast and nowcast, and observing system simulation study for future advanced satellites.



Timothy J. Schmit received the B.S. and M.S. degrees in meteorology from the University of Wisconsin–Madison, Madison, WI, USA, in 1985 and 1987, respectively.

He works at the Advanced Satellite Products Branch (ASPB), NOAA's NESDIS Center of Satellite Applications and Research, Madison. He has been recognized as an international expert in the field of geostationary remote sensing. With bachelor's and master's degrees from the University of Wisconsin–Madison, he has dedicated his career to scientific support of multiple Geostationary Operational Environmental Satellites (GOES) missions and instruments, including the GOES-16/17 Advanced Baseline Imagers. He has authored or coauthored over 100 scientific articles in peer-reviewed journals, several book chapters, and co-edited a book, all associated with some aspect of GOES. His research interests span calibration, visualization, and algorithm development and he is committed to research-to-operations, training others, and science communication.

Dr. Schmit was recently elected as a fellow of the American Meteorological Society. He received several awards, including the Department of Commerce Gold Medal, the National Weather Association T. Theodore Fujita Research Achievement Award, and the 2018 finalist for the Samuel J. Heyman Service to America Award.



Jinlong Li received the B.S. degree in atmospheric dynamics from the Nanjing Institute of Meteorology, Nanjing, Jiangsu, China, in 1985, the M.S. degree in atmospheric dynamics from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, in 1990, and the Ph.D. degree in atmospheric sciences from the Georgia Institute of Technology, Atlanta, GA, USA, in 2001.

Since 2005, he has been working with the Cooperative Institute for Meteorological Satellite Studies (CIMSS), University of Wisconsin–Madison, Madison, WI, USA. He has a broad research interest in satellite remote sensing, data assimilation, and numerical weather forecast. His research interests include atmospheric parameter retrievals, satellite data assimilation and application from current and future generations of U.S. geostationary, and polar-orbiting satellite observations.



Jun Li received the B.S. degree in mathematics from Peking University, Beijing, China, in 1987, and the M.S. and Ph.D. degrees in atmospheric science from the Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, in 1990 and 1996, respectively.

He has authored and coauthored more than 200 peer-reviewed journal publications. His research interests include retrieval of geophysical parameters from both geostationary and polar-orbiting satellite measurements, applications of satellite data in nowcasting, weather forecasting and numerical weather prediction, and future observing system simulation and impact assessment.

Dr. Li has been granted permanent principal investigator status, promoted to Distinguished Scientist, and awarded the Chancellor's Award for Excellence in Research by the University of Wisconsin–Madison.



Andrew K. Heidinger received the B.S. degree in mechanical engineering from Purdue University, West Lafayette, IN, USA, in 1992, and the M.S. and Ph.D. degrees in atmospheric science from Colorado State University, Fort Collins, CO, USA, in 1995 and 1998, respectively.

In 1998, he joined the NOAA NESDIS Office of Research and Applications, College Park, MD, USA, as a Physical Scientist. His primary responsibility was the development of cloud product retrievals from the current NOAA operational polar and geostationary imagers. During this time, he developed the clouds from AVHRR extended processing system, which began generating cloud products operationally in 2001. In 2002, he transferred to the NOAA/NESDIS Advanced Satellite Products, Cooperative Institute for Meteorological Satellite Studies (CIMSS), Space Science and Engineering Center, University of Wisconsin–Madison, Wisconsin, MI, USA. After his move to CIMSS, he became interested in using NOAA satellite data for climate work. To this end, he began work to improve the sensor calibration of historic NOAA data and developed the pathfinder atmospheres extended climate dataset. He also developed the Successive Order of Interaction radiative transfer model. Much of his time, at CIMSS, has been devoted to the preparation for the Geostationary Operational Environmental Satellites-R Series (GOES-R) and Joint Polar Satellite System (JPSS) missions. He has authored or coauthored over 100 scientific articles in peer-reviewed journals.

Dr. Heidinger became a member of the National Aeronautics and Space Administration National Polar-Orbiting Partnership Science Team in 2012. He is a member of the WMO SCOPE-CM Project on satellite climate data and the GEWEX Data Analysis Panel and has served on the International Radiation Commission. He was a recipient of three NESDIS Bronze Awards and one NOAA Silver Award. In 2012, he was named NOAA Employee of the Month. In 2003, he served as the Chair for the Visible Infrared Imaging Radiometer Suite Operational Algorithm Team. In 2006, he became the Chair of the GOES-R Cloud Team, and in 2010, he was named the Chair of the JPSS Cloud Team. He has led the development of the majority of the cloud algorithms for GOES-R and JPSS and has pursued state-of-the-art physical algorithms to meet the evolving needs of NOAA's customers. Since 2006, he has also been involved with the Moderate Resolution Imaging Spectroradiometer Science Team. In 2016, he also became the Co-Chair of the International Cloud Working Group, Darmstadt, Germany.

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